



12-MAVZU · I BOB. HTML VA CSS

CSS animatsiyalar va transformatsiyalar

transform (translate, scale, rotate, skew), transition bilan silliq o'tishlar, @keyframes va animation xossalari; interaktiv tugmalar, yuklanish belgisi, kartalar effektlari.

8-sinf · Web Full-stack dasturlash

Bugun nimani o'rganamiz?

- Siljitish, kattalashtirish, burish — **transform**
- Holat o'zgarganda silliq o'tish — **transition**
- O'zi harakatlanuvchi animatsiya — **@keyframes + animation**
- Effektlar: "tirik" tugmalar, yuklanish belgisi, paydo bo'luvchi kartalar

Transform — shaklni o'zgartirish

CSS

```
.t1 { transform: translateX(30px); }      /* o'ngga */
.t2 { transform: translateY(-20px); }     /* tepaga */
.t3 { transform: scale(1.3); }           /* kattalashtirish */
.t5 { transform: rotate(45deg); }        /* burish */
.t6 { transform: skew(15deg); }          /* qiyshaytirish */
.t7 { transform: rotate(-10deg) scale(1.1); } /* birga */
```

Funksiya	Vazifasi
<code>translate(x, y)</code>	Siljitish
<code>scale(n)</code>	1 = asl o'lcham, 1.1 = 10% katta
<code>rotate(deg)</code>	Burish
<code>skew(deg)</code>	Qiyshaytirish

- Boshqa elementlarga **ta'sir qilmaydi**; `transform-origin` — qaysi nuqtadan

Transition — silliq o'tish



Chiroq: oddiy chiroq bir zumda yonadi; kinoteatr chiroqlari asta-sekin. Siz boshlang'ich va oxirgi holatni berasiz — oradagi "kadrlar"ni brauzer chizadi.

CSS

```
.tugma {  
  background: #4f46e5;  
  transition: background-color 0.3s ease, transform 0.3s ease;  
}  
.tugma:hover {  
  background: #db2777;  
  transform: scale(1.1);  
}
```

transition sintaksisi

CSS

```
transition: background-color 0.3s ease 0s;
/*
├── kechikish (delay)
├── tezlik egri chizig'i
├── davomiylik (s, ms)
└── qaysi xossa (all - hammasi) */
```

Egri chiziq	Xatti-harakat
linear	Bir tekis (robotdek)
ease	Sekin → tez → sekin (standart, tabiiy)
ease-in / ease-out	Sekin boshlanadi / sekin tugaydi

⚠ `transition` ni **oddiy holatga** yozing, `:hover` ga emas — aks holda qaytishi keskin bo'ladi

Animation: 1-qadam — @keyframes "ssenariy"

CSS

```
@keyframes sakrash {  
  0%   { transform: translateY(0); }  
  50%  { transform: translateY(-40px); }  
  100% { transform: translateY(0); }  
}
```

/ Ikki bosqich bo'lsa: */*

```
@keyframes paydo {  
  from { opacity: 0; }  
  to   { opacity: 1; }  
}
```

- **transition** — faqat holat o'zgarganda, $A \rightarrow B$
- **animation** — **o'zi boshlanadi**, ko'p bosqichli, cheksiz takrorlanishi mumkin

Animation: 2-qadam — elementga qo'llash

CSS

```
.top { animation: sakrash 1s ease-in-out infinite; }  
/* nomi | davomiylik | tezlik | takror */
```

Xossa	Qiymatlar
animation-duration	2s , 500ms
animation-iteration-count	1 , 3 , infinite
animation-direction	normal reverse alternate
animation-delay	0.5s
animation-fill-mode	forwards — oxirgi holatda qolish

Misol: yuklanish belgisi (spinner)

CSS

```
.spinner {  
  width: 50px; height: 50px;  
  border: 5px solid #e0e7ff;  
  border-top-color: #4f46e5; /* faqat tepa qismi rangli */  
  border-radius: 50%;  
  animation: aylan 0.8s linear infinite;  
}  
  
@keyframes aylan {  
  to { transform: rotate(360deg); }  
}
```

Misol: paydo bo'luvchi kartalar

CSS

```
@keyframes paydo {  
  from { opacity: 0; transform: translateY(30px); }  
  to   { opacity: 1; transform: translateY(0); }  
}  
  
.karta {  
  animation: paydo 0.6s ease both;  
  transition: transform .25s, box-shadow .25s;  
}  
  
.karta:nth-child(2) { animation-delay: 0.15s; }  
.karta:nth-child(3) { animation-delay: 0.3s; }  
.karta:hover {  
  transform: translateY(-8px);  
  box-shadow: 0 16px 30px rgba(79,70,229,.25);  
}
```

Misol: porlovchi tugma

CSS

```
.porla {  
  background: linear-gradient(90deg, #4f46e5, #db2777, #4f46e5);  
  background-size: 200% 100%;  
  border-radius: 999px;  
  animation: oqim 3s linear infinite;  
  transition: transform .2s;  
}  
.porla:hover { transform: scale(1.06); }  
.porla:active { transform: scale(.96); }  
@keyframes oqim { to { background-position: -200% 0; } }
```

Unumdorlik va foydalanuvchilar

- Eng tez xossalar: `transform` va `opacity`
- `width`, `top`, `margin` animatsiyasi sahifani sekinlashtiradi
- **Me'yorni** saqlang — hamma narsa sakrasa, ko'z charchaydi

CSS

```
@media (prefers-reduced-motion: reduce) {  
  * { animation: none !important; transition: none !important; }  
}
```

- Ba'zi odamlarda harakatdan bosh aylanadi — ularni hurmat qiling

Amaliyot: portfolio jonlanadi

- 1 Barcha tugma va havolalarga silliq `:hover` (`transition`)
- 2 Loyiha kartalari hover'da **8px ko'tarilsin**, soya kuchaysin
- 3 Hero sarlavha va tugma pastdan "**paydo**" bo'lsin (kechikish bilan)
- 4 Profil rasmiga "**nafas olish**": `scale(1) → 1.03 → 1`, 4s, cheksiz
- 5 🌞 Ijodiy: faqat CSS bilan "**Quyosh tizimi**" — 2 sayyora turli tezlikda aylanadi

Ko'p uchraydigan xatolar

- ❌ Ikkita `transform` qatori — ikkinchisi birinchisini o'chiradi; bittada yozing:
`rotate(10deg) scale(1.2)`
- ❌ `transition: all` hamma joyda — kutilmagan xossalar ham animatsiya bo'ladi
- ❌ `@keyframes aylanish` va `animation: aylanis` — nomlar mos emas
- ❌ `display: none` ni animatsiya qilish — `opacity` va `visibility` ishlating

Savol

Elementni 1.2 barobar kattalashtiruvchi kod?

A `transform: size(1.2)`

B `zoom: 120px`

C `transform: scale(1.2)`

D `transform: grow(20%)`

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Hover effekti silliq bo'lishi uchun `transition` qayerga yoziladi?

A Elementning oddiy holatiga

B Faqat `:hover` ga

C `@keyframes` ichiga

D `body` ga

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Animatsiyani cheksiz takrorlash uchun qaysi qiymat?

A repeat

B loop

C forever

D infinite

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Qaysi xossalarni animatsiya qilish eng tez (unumdor)?

A width va height

B transform va opacity

C margin va padding

D top va left

Javobni tanlang — to'g'ri variant yashil bo'ladi



Uyga vazifa

- 1 "Quyosh tizimi"ni yakunlang
- 2 3 xil spinner: aylanuvchi doira, sakrovchi 3 nuqta, to'lib boruvchi chiziq
- 3 **Flip card:** kartani 180° aylantirib orqa tomonini ko'rsatish (`rotateY` , `backface-visibility`)



Yodda tuting

- `transform`: translate, scale, rotate, skew
- `transition: xossa vaqt tezlik;` — oddiy holatga yoziladi
- `@keyframes nom { 0% {} 100% {} }` + `animation: nom 2s infinite;`
- Unumdorlik: `transform` va `opacity`
- Me'yorni saqlang, `prefers-reduced-motion` ni hurmat qiling



Savollar bormi?

Keyingi mavzu: **13. Jamoa bilan ishlash va loyiha qilish. Netlify va Vercel'ga deploy**

Batafsil matn, misollar va testlar: **12-css-animatsiya.html** darsida